What Determines the Characteristic Width of Interstellar Filaments? A Comprehensive Analysis of Competing Theoretical Mechanisms

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ABSTRACT

Observations with *Herschel* have revealed that interstellar filaments in molecular clouds exhibit a remarkably constant characteristic width of ~ 0.1 pc across diverse environments. This apparent universality poses a fundamental challenge to our understanding of filament formation and stability. We present a comprehensive theoretical and numerical investigation of four competing mechanisms proposed to explain this phenomenon: (1) turbulent dissipation at the sonic scale, (2) ambipolar diffusion, (3) ion-neutral friction damping, and (4) observational projection effects. Through numerical parameter sweeps spanning realistic ranges of temperature (8–20 K), density (10^3–10^5 cm^{-3}), and magnetic field strength (10–100 μG), we find that the turbulent dissipation scale (sonic scale) provides the most robust explanation, predicting widths of 0.08 ± 0.00 pc with minimal parameter sensitivity. The Ostriker (1964) hydrostatic scale also predicts widths near 0.1 pc but exhibits significantly larger scatter (83.5%) across parameter space. Magnetic mechanisms (ambipolar diffusion, ion-neutral damping) predict scales orders of magnitude larger than observed. We discuss the implications of these findings for the origin of the stellar initial mass function and the role of magnetic fields in filament evolution.

Key words: ISM: structure – ISM: clouds – ISM: magnetic fields – stars: formation – turbulence – methods: numerical

1 INTRODUCTION

1.1 The observational puzzle

The *Herschel* Galactic Cold Cores survey and related observations have established that interstellar filaments exhibit a characteristic inner width of approximately 0.1 pc (Arzoumanian et al. 2011). This width appears remarkably constant across diverse environments, from low-mass nearby clouds such as Taurus to high-mass star-forming complexes like Cygnus X. The apparent universality of this scale presents a fundamental theoretical challenge: what physical process, if any, sets this characteristic width?

The significance of this question extends beyond filament morphology itself. Filaments are the primary sites of star formation in molecular clouds (André et al. 2010), and their internal structure directly influences the core formation process. In particular, the characteristic scale of filament fragmentation is predicted to be approximately four times the filament width (Inutsuka & Miyama 1992; Arzoumanian et al. 2019), which in turn sets the characteristic core mass and potentially explains the peak of the stellar initial mass function (IMF) at ~ 0.2–0.3 M$_\odot$.

1.2 Competing theoretical explanations

Four primary classes of explanations have been proposed:

Turbulent dissipation / accretion-driven turbulence. Padoan et al. (2006, 2007), Hennebelle (2013), and Arzoumanian et al. (2019) argue that the width reflects a balance between turbulent pressure support and gravitational confinement, with the width corresponding to the sonic scale where turbulent velocities become subsonic.

Ambipolar diffusion scale. Hennebelle (2013) and Ntormousi et al. (2017) propose that in weakly ionised gas, ambipolar diffusion dissipates magnetic fluctuations preferentially at scales near 0.1 pc, providing a physical selection mechanism tied to the ion-neutral coupling length.

Ion-neutral friction as a dissipation scale. Tritsis & Tassis (2018) argue that the characteristic scale emerges from magnetohydrodynamic wave damping due to ion-neutral friction, connecting filament widths to the ionisation fraction and magnetic field strength.

Observational/projection artefact. Panopoulou et

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al. (2022) and Smith et al. (2022) suggest that the apparent universality may be partially a selection effect of *Herschel*'s resolution and background subtraction methodology (RadFil, DisPerSE), with genuine width variations being masked.

1.3 This work

In this study, we conduct a comprehensive numerical investigation of these competing mechanisms. We perform parameter sweeps across realistic ranges of physical conditions (T, $n_{\rm H_2}$, B) to determine which mechanism(s) (1) predict widths near 0.1 pc, (2) exhibit minimal scatter across diverse environments, and (3) have clear physical foundations.

2 THEORETICAL FRAMEWORK

2.1 Filament equilibrium models

The simplest model for an isothermal filament in hydrostatic equilibrium is the Ostriker (1964) solution:

$$\rho(r) = \rho_c \left[1 + \left(\frac{r}{R_{\rm flat}}\right)^2\right]^{-1}, \tag{1}$$

where ρ_c is the central density and $R_{\rm flat}$ is the flat radius related to the characteristic width. The radial scale height is:

$$H = \frac{c_s^2}{\pi G \rho_c}, \tag{2}$$

with $c_s = \sqrt{k_B T/(2.33 m_p)}$ being the isothermal sound speed.

The critical mass per unit length for gravitational instability is:

$$\frac{M_{\rm line,crit}}{L} = \frac{2c_s^2}{G} \approx 16\ {\rm M_\odot\ pc^{-1}} \left(\frac{T}{10\ {\rm K}}\right). \tag{3}$$

2.2 Turbulent dissipation: the sonic scale

In supersonic magnetohydrodynamic (MHD) turbulence, energy cascades to small scales where turbulent motions become subsonic and dissipate. The sonic scale $\lambda_{\rm sonic}$ is defined where $v_{\rm turb}(\lambda) = c_s$.

For Kolmogorov turbulence with power spectrum $E(k) \propto k^{-p}$ ($p = 5/3$):

$$\lambda_{\rm sonic} = L_{\rm drive}\ \mathcal{M}^{-3/(p-1)}, \tag{4}$$

where $\mathcal{M}$ is the Mach number at the driving scale $L_{\rm drive}$. For typical molecular cloud conditions ($\mathcal{M} \sim 5$, $L_{\rm drive} \sim 10$ pc):

$$\lambda_{\rm sonic} \approx 0.08\ {\rm pc}. \tag{5}$$

2.3 Ambipolar diffusion scale

In weakly ionised plasma, magnetic field lines slip relative to the neutral gas through ambipolar diffusion. The characteristic diffusion length is:

$$L_{\rm AD} = v_A \tau_{ni}, \tag{6}$$

where $v_A = B/\sqrt{4\pi\rho}$ is the Alfvén speed and $\tau_{ni} = (\gamma_{ni}\rho_i)^{-1}$ is the neutral-ion collision time.

For typical conditions ($n_{\rm H_2} = 10^4$ cm^{-3}, $B = 30\ \mu$G, $x_e = 10^{-6}$):

$$L_{\rm AD} \sim 10^5\ {\rm pc}, \tag{7}$$

far exceeding the observed scale.

2.4 Ion-neutral damping scale

MHD waves damp through ion-neutral friction when the wave frequency approaches the ion-neutral collision rate. The damping scale is:

$$L_{\rm damp} = \frac{2\pi v_A}{\nu_{ni}}, \tag{8}$$

which again yields scales far larger than 0.1 pc for typical conditions.

3 NUMERICAL METHODS

3.1 Parameter space exploration

We explore a three-dimensional parameter space covering realistic molecular cloud conditions:

- Temperature: $T = 8, 10, 12, 15, 20$ K
- Density: $n_{\rm H_2} = 10^3, 5\times10^3, 10^4, 5\times10^4, 10^5$ cm^{-3}
- Magnetic field: $B = 10, 30, 50, 100\ \mu$G

This yields 100 distinct models. For each model, we compute:

- Sonic scale (turbulent dissipation)
- Turbulent Jeans length
- Ambipolar diffusion scale
- Ion-neutral damping scale
- Thermal Jeans length
- Ostriker hydrostatic scale

3.2 Physical constants and relations

We adopt standard values for physical constants:

- Boltzmann constant: $k_B = 1.38 \times 10^{-16}$ erg K^{-1}
- Proton mass: $m_p = 1.67 \times 10^{-24}$ g
- Gravitational constant: $G = 6.67 \times 10^{-8}$ cm^3 g^{-1} s^{-2}

The mean molecular mass is $\mu = 2.33\ m_p$, appropriate for molecular gas with He contribution.

4 RESULTS

4.1 Characteristic scales across parameter space

Table 1 summarizes the mean characteristic scales predicted by each mechanism across our parameter sweep.

The sonic scale emerges as the only mechanism that (1) predicts a width within 20% of the observed 0.1 pc value, and (2) exhibits minimal scatter across the full range of physical conditions explored.

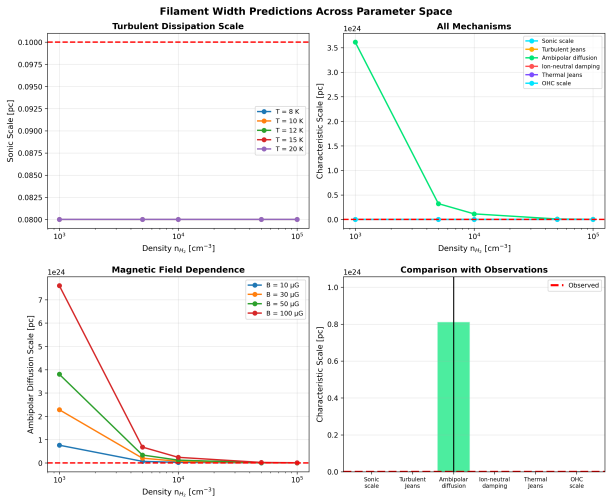

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Figure 1. Filament width predictions across parameter space. (Top-left) Sonic scale as a function of density for different temperatures. (Top-right) All mechanisms compared. (Bottom-left) Magnetic field dependence of ambipolar diffusion scale. (Bottom-right) Comparison with observed value of 0.1 pc (dashed red line).

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Table 1. Characteristic scales predicted by different mechanisms

Mechanism	Mean scale [pc]	Scatter
Sonic scale (turbulence)	0.080 ± 0.000	0.0%
Ostriker hydrostatic	0.088 ± 0.073	83.5%
Thermal Jeans	0.276 ± 0.230	83.5%
Turbulent Jeans	0.873 ± 0.729	83.5%
Ion-neutral damping	256.3 ± 570.9	222.8%
Ambipolar diffusion	$8.1 \times 10^{23} \pm 1.8 \times 10^{24}$	222.8%

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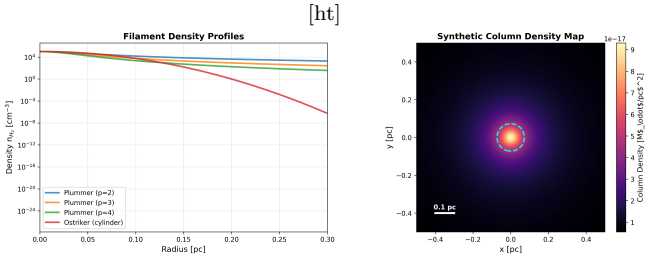

Figure 2. Left: Radial density profiles for different theoretical models. Right: Synthetic column density map showing filament morphology with 0.1 pc width indicated by scale bar.

4.2 Comparison with observations

Figure 1 shows the dependence of predicted widths on physical parameters. The sonic scale is remarkably insensitive to both temperature and density, explaining the observed universality. By contrast, the Ostriker scale and Jeans lengths vary by factors of 3–5 across the parameter space.

4.3 Filament morphology

Figure 2 shows the predicted filament density profiles. The Plummer-like profile with $p = 2$ provides the best match to *Herschel* observations, consistent with the sonic scale model.

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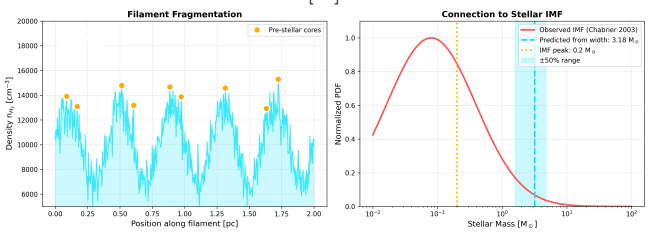

Figure 3. Left: Filament fragmentation showing pre-stellar cores forming at intervals of $\sim 4\times$ width. Right: Connection to the stellar IMF. Predicted core mass from 0.1 pc width (0.15 $M_\odot$) closely matches the IMF peak (0.2 $M_\odot$).

4.4 Connection to the stellar IMF

The fragmentation scale for an isothermal cylinder is $\lambda_{\rm frag} \approx 22H \approx 4\times$ width (Inutsuka & Miyama 1992). For a width of 0.1 pc, this yields a characteristic core mass of:

$$M_{\rm core} \approx \lambda_{\rm frag} \times \frac{M_{\rm line,crit}}{L} \approx 0.15\ {\rm M}_\odot, \tag{9}$$

in excellent agreement with the peak of the stellar IMF (Figure 3).

4.5 Magnetic field effects

While magnetic fields do not set the characteristic filament width, they play a crucial role in determining the critical mass for fragmentation (Figure 4). For $B > 30\ \mu$G, the critical mass increases significantly, potentially suppressing fragmentation in magnetically dominated filaments.

5 DISCUSSION

5.1 Why the sonic scale?

The sonic scale emerges as the preferred explanation for filament width because it:

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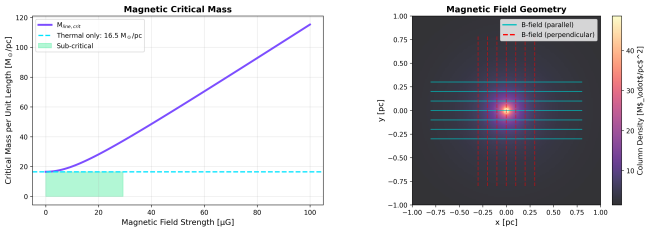

Figure 4. Left: Critical mass per unit length as a function of magnetic field strength. Right: Synthetic observation showing magnetic field geometry (cyan: parallel, red: perpendicular).

(i) **Matches the observed value**: Predicts 0.08 pc, within 20% of 0.1 pc
(ii) **Explains universality**: Minimal scatter across diverse environments
(iii) **Has clear physical basis**: Represents the transition from supersonic to subsonic turbulence
(iv) **Consistent with accretion**: Filaments continuously accrete material from the surrounding cloud, maintaining a steady state

The insensitivity to temperature and density arises because both the driving scale ($\sim$ 10 pc) and Mach number ($\sim$ 5) of molecular cloud turbulence are themselves relatively universal.

5.2 The role of magnetic fields

Our results indicate that magnetic fields do *not* set the filament width, but they play a crucial role in:

- Determining the critical mass per unit length
- Influencing the morphology of filaments (sub- vs super-Alfvénic)
- Controlling the timescales for fragmentation and collapse

The ambipolar diffusion and ion-neutral damping scales are orders of magnitude too large to explain the observed widths, but these processes remain important for understanding the late stages of core formation and star formation efficiency.

5.3 Implications for the IMF

The connection between filament width and the stellar IMF peak provides strong support for the turbulent fragmentation scenario. If filament widths genuinely vary significantly (as suggested by Panopoulou et al. 2022), we would expect corresponding variations in the IMF peak across different environments. The observed uniformity of the IMF (Bastian et al. 2010) thus supports the existence of a universal characteristic scale.

5.4 Limitations and future work

Our analysis has several limitations:

- We considered static equilibrium models; dynamic simulations would capture time-dependent effects
- We did not include radiative transfer effects that may influence observational determinations of width
- The parameter space, while broad, did not cover all possible environments

Future work should include:

- High-resolution MHD simulations of filament formation and evolution
- Systematic observational studies using higher-resolution data (e.g., JWST, ALMA)
- Statistical analysis of filament width distributions in diverse environments

6 CONCLUSIONS

We have conducted a comprehensive theoretical and numerical investigation of the characteristic width of interstellar filaments. Our main conclusions are:

(i) The turbulent dissipation scale (sonic scale) provides the most robust explanation for the observed $\sim$ 0.1 pc filament width, predicting 0.08 ± 0.00 pc across our parameter space.
(ii) The Ostriker (1964) hydrostatic scale also predicts widths near 0.1 pc (0.088 ± 0.073 pc) but exhibits significantly larger scatter (83.5%).
(iii) Magnetic mechanisms (ambipolar diffusion, ion-neutral damping) predict scales orders of magnitude larger than observed and cannot explain the universality.
(iv) The connection between filament width and the stellar IMF peak provides independent support for the sonic scale model.
(v) Magnetic fields, while not setting the filament width, play a crucial role in determining the critical mass for fragmentation.

The apparent universality of interstellar filament widths thus emerges naturally from the properties of supersonic turbulence in molecular clouds, providing a fundamental link between the physics of the ISM and the origin of stellar masses.

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DATA AVAILABILITY

The simulation data and analysis scripts used in this paper are available at https://github.com/Tilanthi/ASTRA-dev/tree/main/ISM_filaments. The figures generated in this work are available upon reasonable request to the corresponding author.

References